

RAGA SIRI POTLURI

Data & Generative AI Engineer | Multi-Cloud Data Platforms (Azure, AWS, GCP) | Azure OpenAI, ETL, RAG, LLM's, Bedrock, Databricks, AWS Glue | Healthcare, Finance, Retail, Big Data Platforms
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PROFESSIONAL SUMMARY

- **Data and Generative AI Engineer** with around **5 years** of experience designing enterprise-scale **ETL pipelines** and **RAG/LLM** architectures across **AWS, Azure, and GCP**, delivering governed analytics and AI-driven solutions supporting 10+ business teams in **finance, retail, and healthcare** domains.
- Proven expertise building large-scale **data lakes** and **data warehouses** using **AWS S3, Azure Data Lake Storage, and BigQuery**, processing over 1B+ records and 6TB daily to support analytics, reporting, **RAG** architectures, and **LLM**-driven decision systems.
- Strong background in credit risk and delinquency analytics, delivering **ETL** pipelines on **AWS Glue** and **AWS Redshift** handling 25M monthly loan records across 12 products and 3K branches for regulatory, **MIS** reporting, and **AI**-powered risk insights.
- Hands-on experience designing and implementing **RAG/LLM** architectures with **Azure Databricks, Azure OpenAI, AWS Bedrock, and Azure Cognitive Search**, enabling semantic search, embeddings, and grounded summarization across 18M healthcare documents serving 15K daily users.
- Expertise in **Spark** and **PySpark** for distributed batch processing, implementing 120M text chunk pipelines, embedding generation workflows, and 40+ complex metrics to support forecasting, risk analytics, and large-scale **GenAI** workloads.
- Delivered supply chain and replenishment analytics on **GCP** using **Dataflow, BigQuery, and Cloud Composer**, orchestrating 25 pipelines and supporting 2K retail locations with **SLA**-driven inventory insights.
- Strong focus on data quality and governance, implementing 100+ validation rules and **RBAC** controls using **AWS IAM** and **Azure Active Directory**, supporting 300+ users while ensuring audit-ready, secure, and compliant data access for analytics and **LLM** applications.
- Experienced in **CI/CD** pipeline orchestration and observability using **AWS CloudWatch, Azure Monitor, Cloud Logging, and Azure DevOps**, tracking 50+ operational metrics to ensure reliability, freshness, and performance across enterprise data and **RAG** platforms.
- Demonstrated ability to optimize performance at scale, tuning **BigQuery** and **AWS Redshift** tables exceeding 1B rows and optimizing **Azure Databricks** jobs across 5 clusters to support demanding analytics and **LLM** workloads.
- Collaborative engineer with experience partnering with 20+ stakeholders including analysts, clinicians, and product teams, translating business rules into scalable data engineering, **RAG** architectures, and **LLM**-powered **Generative AI** solutions across 4+ regulated industries.

TECHNICAL SKILLS

Programming & Query Languages: Python, SQL, PySpark

Cloud Platforms: AWS, Azure, GCP

AWS Services: AWS S3, AWS Glue, AWS Lambda, Redshift, AWS Bedrock, EMR, CloudWatch, OpenSearch, IAM, KMS, CodePipeline, CloudFormation

Azure Services: Azure Data Factory, Azure Databricks, Azure Data Lake Storage Gen2, Azure Synapse, Azure Monitor, Azure Purview, Power BI

GCP Services: BigQuery, Dataflow, Cloud Composer, Cloud Storage, Pub/Sub, Dataplex, Cloud Logging, Cloud Monitoring, Cloud Build

Data Engineering & Analytics: ETL, ELT, DBT, Data Lake, Data Warehouse, Lakehouse, Batch Processing, Streaming Pipelines, Data Modeling, Data Quality, Data Validation, Data Lineage, Metadata Management

Big Data & Processing: Spark, Delta Lake, BigQuery SQL, Distributed Processing, Partitioning, Clustering, Performance Tuning

Generative AI & ML: Gen AI, LLMs, RAG, Vector Search, Embeddings, Prompt Engineering, Semantic Search, Model Evaluation

Databases & Storage: BigQuery, Redshift, Snowflake, Delta Lake, OpenSearch, Cloud Storage, Azure Data Lake, Amazon S3

Orchestration & DevOps: Cloud Composer, Azure DevOps, AWS CodePipeline, CI/CD, Git, Infrastructure as Code
Monitoring, Alerting

Security & Governance: IAM, RBAC, KMS, VPC Service Controls, GxP Compliance, HIPAA, Regulatory Reporting, Audit Readiness

PROFESSIONAL EXPERIENCE

Data and Generative AI Engineer | Optum | Dallas, Texas

Jan 2025 - Present

Project: Built a secure, **Azure-native Generative AI assistant** enabling natural-language search, summarization, and analytics across **healthcare claims, policy, and clinical documents** using **Azure Databricks, Azure Data Lake Storage Gen2, Azure OpenAI, and Azure Cognitive Search**.

- Designed end-to-end **ETL and RAG pipelines** on **Azure Databricks** to ingest and normalize 18M healthcare documents from six source systems into **Azure Data Lake Storage Gen2**, enabling scalable semantic search and analytics-ready datasets.
- Implemented large-scale document preprocessing using **Spark** and **SQL** on **Azure Databricks**, generating 120M text chunks with optimized token windows and overlap, improving retrieval accuracy across 25 claims and policy domains.
- Built vector indexing and enrichment workflows using **Azure Databricks** and **Azure Cognitive Search**, storing 120M embeddings and enabling sub-second semantic queries across 3TB of unstructured healthcare content.
- Integrated **Azure OpenAI models** into Databricks-driven RAG pipelines to deliver grounded responses for 15K daily enterprise queries, supporting claims adjudication, policy interpretation, and operational research across 10 business teams.
- Developed **SQL-based metadata filtering logic** combining vector similarity with structured attributes, significantly improving precision for clinical, claims, and compliance analytics use cases.
- Orchestrated 12 production-grade **ETL workflows** using **Azure Data Factory**, ensuring daily ingestion of 2M new documents and embeddings with end-to-end SLAs under 90 minutes for downstream GenAI consumption.
- Implemented **CI/CD pipelines** using **Azure DevOps** to automate deployment of **Azure Databricks notebooks, Azure Data Factory pipelines, and SQL artifacts** across 3 environments, reducing manual releases and cutting promotion time by 40%.
- Enforced data quality and governance by implementing 80+ validation rules using **Spark** and **SQL**, covering completeness, schema drift, and duplicate detection across 2M daily ingested records in regulated healthcare workflows.
- Enabled secure and compliant access through **Azure Active Directory, RBAC**, and service principals, supporting 300+ internal users while ensuring HIPAA-aligned handling of PHI across analytics and GenAI workloads.
- Optimized **Azure Databricks clusters** and **Spark jobs** processing 6TB daily, reducing embedding generation runtimes by 32% while maintaining throughput for peak enterprise GenAI usage.
- Implemented monitoring and alerting using **Azure Monitor** and **Log Analytics**, tracking 40+ metrics including query latency, pipeline failures, retrieval depth, and model response errors for proactive tuning.
- Delivered **Power BI dashboards** sourced from curated **Azure Databricks** and **SQL datasets** to provide visibility into document coverage, query trends, and pipeline SLAs for claims, compliance, and operations leadership.

- Partnered with **clinicians, claims analysts, and compliance teams** to translate business rules into ETL logic, SQL transformations, and prompt refinements, aligning **Azure-based GenAI** outputs with operational definitions across 4 healthcare programs.

Data Engineer | Sam's Club | Bentonville, Arkansas

May 2024 – Dec 2024

Project: Built **AWS**-native supply chain and replenishment data pipelines processing 120M+ inventory, logistics, and supplier records monthly, using **Amazon S3, AWS Glue, and Redshift**, and integrated **Bedrock** based **RAG** and **LLM** workflows to deliver **SLA**-driven insights for 2K clubs and 15 distribution centers.

- Architected end-to-end **ETL** pipelines on **AWS Glue** and **Amazon S3** to ingest and transform 120M supplier, shipment, and store inventory records monthly into **Amazon Redshift**, enabling unified supply chain analytics across 2K retail clubs and 15 distribution centers.
- Implemented real-time ingestion using **Amazon Kinesis** and **AWS Lambda**, enabling incremental inventory updates and streaming transformations that improved replenishment responsiveness and operational visibility.
- Built curated **warehouse layers** in **Amazon Redshift** using advanced **SQL** transformations, standardizing schemas across 35 product categories and modeling 50+ SKU-location attributes to support lead-time and inventory movement analytics.
- Engineered scalable batch processing in **AWS Glue** and **PySpark** to compute 40+ supply chain **KPIs** across 18M **SKUs**, ensuring consistent, analytics-ready datasets for forecasting, operations reporting, and downstream AI use cases.
- Orchestrated 25 interdependent workflows using **AWS Step Functions** and **CloudWatch**, coordinating **S3** loads, streaming ingestion, and **Redshift** transformations with **SLA** tracking under 75 minutes from data arrival.
- Designed and optimized fact and dimension models in **Amazon Redshift**, tuning distribution and sort keys on tables exceeding 900M rows, reducing query latency by 32% across 30 recurring operational dashboards.
- Implemented data governance using **AWS Glue** Data Catalog, **IAM**, and **KMS**, enabling end-to-end lineage, role-based access, and encryption controls across **S3, Glue, and Redshift** assets supporting 40+ users.
- Developed data quality frameworks in **PySpark** and **SQL** enforcing 120+ business validation rules across 120M rows per run, improving trust and reliability of replenishment and logistics reporting datasets.
- Maintained 24 rolling monthly snapshots in **Amazon S3** and **Redshift** to support seasonal trend analysis, historical comparisons, and back-testing of replenishment forecasting logic across 6 merchandise divisions.
- Integrated curated **Redshift** and **S3** datasets with **AWS Bedrock** foundation models to enable enterprise **RAG pipelines**, combining structured supply chain data with contextual retrieval to generate grounded **LLM**-based operational insights.
- Designed embedding and retrieval workflows using **Amazon OpenSearch** for vector indexing, enabling semantic search across supplier policies and logistics documents to support **Bedrock**-powered decision assistance.
- Implemented prompt engineering, context filtering, and response grounding strategies within **Bedrock**-hosted **LLM workflows**, ensuring accurate, citation-backed summaries for supply chain planners and operations teams.
- Automated **CI/CD** using **AWS CodePipeline** and **CloudFormation** to deploy **Glue** jobs, Step Functions workflows, **Redshift** artifacts, and **Bedrock** integration components across environments with zero production rollbacks.
- Integrated monitoring and observability using **Amazon CloudWatch**, tracking 60+ pipeline and model metrics, enabling proactive alerts for data delays, schema drift, and AI response latency issues.
- Collaborated with 9 supply chain analysts and domain stakeholders to translate replenishment logic into governed **AWS** data transformations and **RAG-driven AI** capabilities aligned with enterprise decision standards.

Project: Built a **GCP-native credit risk and delinquency analytics platform** processing high-volume lending data. Delivered governed, secure **ETL** pipelines on **Cloud Storage**, **Dataflow**, and **BigQuery**, enabling portfolio risk tracking, early warning signals, and RBI-aligned MIS reporting at scale.

- Designed end-to-end **ETL** pipelines on **Dataflow** to ingest and transform 25M loan repayment and installment records per month from 5 core lending systems into **Cloud Storage**, enabling centralized credit risk and delinquency analytics.
- Built **Cloud Storage** data lake layers with standardized schemas covering 12 loan products and 3K branches, supporting loan performance, overdue tracking, and 24 months of historical trend analysis.
- Implemented scalable transformations using **Dataflow** and **PySpark** to compute 30+ delinquency indicators across 18M active accounts, ensuring consistent downstream risk metrics.
- Modeled analytics-ready fact and dimension tables in **BigQuery** with 40+ attributes per loan, enabling sub-second queries for risk dashboards, portfolio reviews, and executive **MIS** across 10 business teams.
- Automated daily batch orchestration using **Cloud Functions** and **Cloud Monitoring**, coordinating 15 dependent pipelines with **SLA** monitoring and delivering datasets within 90 minutes of source data arrival.
- Implemented comprehensive **data quality frameworks** enforcing 100+ business rules across 25M rows per run, supporting audit readiness, regulatory reporting, and analytics trust.
- Maintained 36 monthly rolling snapshots in **Cloud Storage** to enable portfolio aging, cohort analysis, and risk back-testing across 8 lending segments.
- Optimized **BigQuery** performance by tuning partitioning and clustering on tables exceeding 1.2B rows, improving analytics response times by 35% across 20 recurring reports.
- Integrated **Cloud Monitoring** logs and metrics to monitor 50+ pipeline health indicators, enabling proactive alerts for delayed feeds, schema drift, and data volume anomalies across 15 batch workflows.
- Delivered RBI-aligned **MIS** datasets refreshed twice daily, supporting 6 regulatory and finance reporting processes with zero manual intervention.
- Enforced enterprise-grade security using **GCP IAM** and **Cloud KMS**, ensuring least-privilege access and encryption of sensitive financial data across 12 roles and 30+ internal consumers.
- Implemented **CI/CD** and infrastructure governance using **Cloud Build** and **Deployment Manager**, enabling repeatable deployment of **Dataflow** jobs, IAM policies, and monitoring resources.
- Collaborated with 7 risk analysts and product stakeholders to translate credit policy rules into governed GCP data transformations across 4 lending business units.

CERTIFICATION

- AWS Certified Data Engineer - Associate (DEA-C01)
- Microsoft Certified: Azure Data Engineer Associate (DP -203)
- Microsoft Certified: Azure AI Engineer Associate (AI -102)
- Google Professional Data Engineer (GCP-PDE)
- Databricks Certified Data Engineer Associate (PR000054)

EDUCATION

The University of Texas at Dallas - Richardson, USA

Master of Science in Computer Science

GPA: 3.6/4